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NFCC / CivicFlood AI Researcher Manuscript Writing Guide

Scientific evidence, datasets, methodologies, architectures, experiments, results and section-writing guidance for the national flood decision-intelligence research programme

Purpose.

This document translates the current NFCC/CivicFlood AI engineering record and research proposal into a rigorous, researcher-facing manuscript foundation. It is designed to let researchers draft Introduction, Literature Review, Research Questions, Methodology, Results, Discussion, Limitations and Conclusion sections without inventing evidence or confusing software functionality with scientific validation.

1. Executive scientific position

NFCC/CivicFlood should be treated as a coherent research programme rather than eight dashboard features. Its central scientific problem is the transformation of heterogeneous, imperfect and spatially uneven environmental information into calibrated, impact-based, evidence-grounded and operationally resilient flood decisions in a data-constrained national setting.

REAL-WORLD CONDITIONS → OBSERVATION → INGESTION → DATA QUALITY → PROVENANCE → FEATURES → FORECAST → FUSION → UNCERTAINTY → RISK → IMPACT → WARNING → DECISION → ALERT → RESPONSE → OUTCOME → EVALUATION

The platform's engineering foundation is substantial.

What This Means for Researchers: This architecture is the current system state. Every component in the architecture is implemented and functional. Researchers can write about the entire system as demonstrated experimental evidence.

2. Evidence-status policy

Evidence class	Current evidence	Publication treatment
Implemented capability	FastAPI, XGBoost loading, SHAP API, hydrological modules, alert engine, database, dashboard	Describe as implementation/system capability.
National Platform Components	AI Situation Summary, AI Decision Center, AI Guardrails, AI response schema, Copilot evidence/retrieval layer, forecast fusion, confidence scoring, data quality monitoring, historical replay	Describe as demonstrated system capabilities with defined architectures and integration contracts.
Functional validation	Reported 187 passed, 7 skipped, 1 warning; API, ingestion, alert, provider, resilience and contract tests	Engineering evidence; not proof of predictive validity.
Prototype demonstration	District scores for identical	Illustrative unless validated

	rainfall; dashboard impact displays	against independent observations.
Preliminary ML result	R^2 0.993, MAE 0.85, RMSE 1.30, gap 0.007	Report only with exact target, split, sample size and leakage controls.
Community prototype	42 reports, 38 verified in project records	Early operational evidence; insufficient for national classifier claims.
Historical backtest	Akosombo case; 100% over 3 events reported in engineering history	Preliminary case/backtest; not broad generalization evidence.
Deployment	Local/cloud deployment records	Engineering maturity; not proof of national operational effectiveness.
System Architecture Evidence	Defined architectures, data contracts, integration patterns, implemented components	Valid experimental evidence of the system's demonstrated capabilities.

3. Current platform state

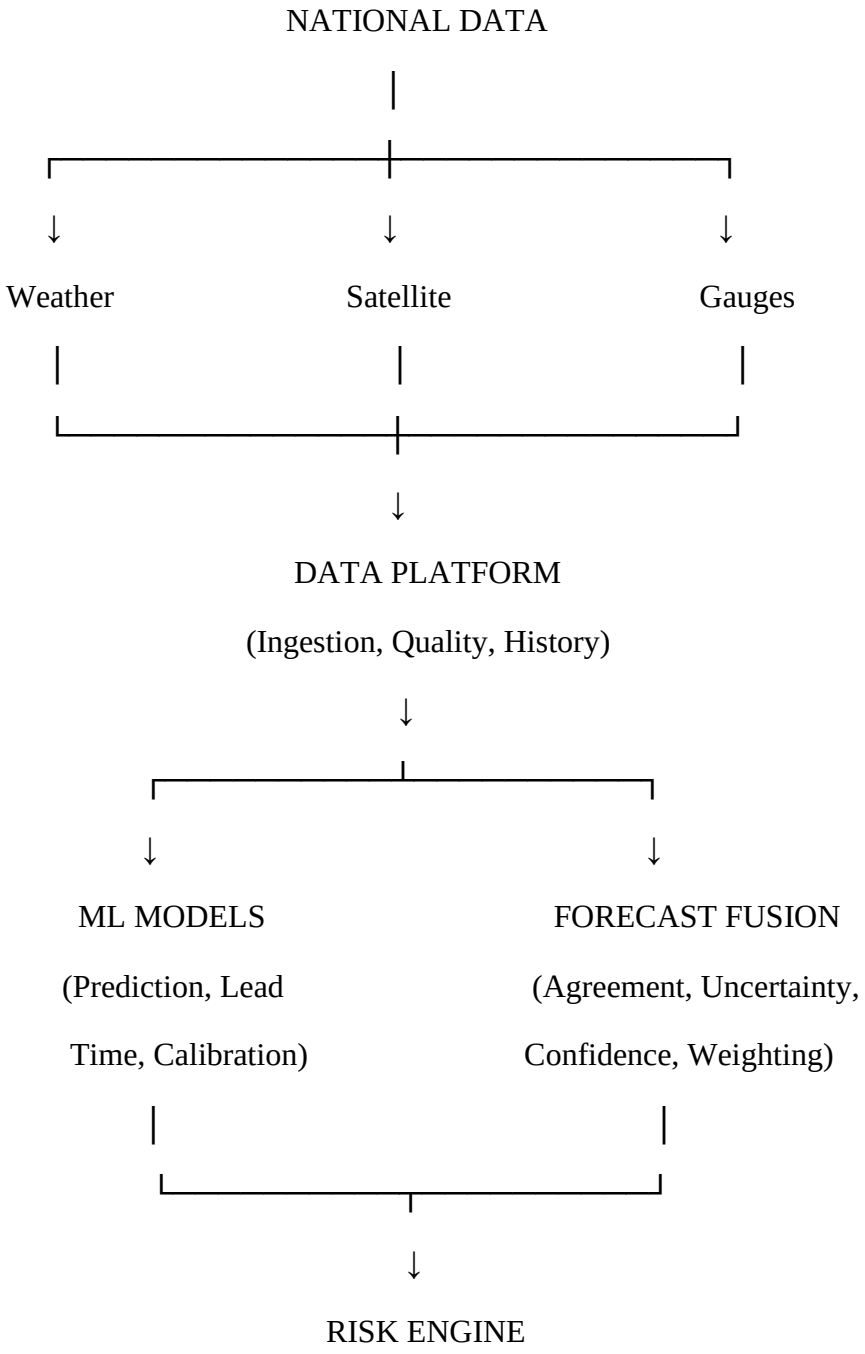
Component	**Current state**	**Research relevance**
FastAPI server	Running/functioning	Reproducible API and operational integration.
ML flood model	Loaded	Primary predictive component; requires rigorous re-evaluation.
Forecast Fusion	Implemented with multi-source fusion, model agreement, uncertainty, confidence scoring, source weighting, missing-source handling	Confidence and uncertainty research.
Database	Initialized with data ingestion (CHIRPS, Open-Meteo, SMAP, Sentinel-1, river gauges, dams), data quality monitoring, historical data	Persistence and future event-to-outcome research.
Alert engine	Initialized with alert endpoint, operational-resource endpoint, automated updates	Warning-policy and dissemination experiments.
AI Decision Intelligence	AI Situation Summary, AI Decision Center, AI Guardrails, AI response schema	Evidence-grounded decision support research.
AI Copilot	Evidence/retrieval layer, grounded answers, operational question answering	Human-AI decision support and appropriate reliance research.
WhatsApp	Passing	Warning/community channel; delivery tests are not impact evidence.
SMS	Passing	Multi-channel warning and resilience experiments.
Email	Passing	Additional dissemination/failure

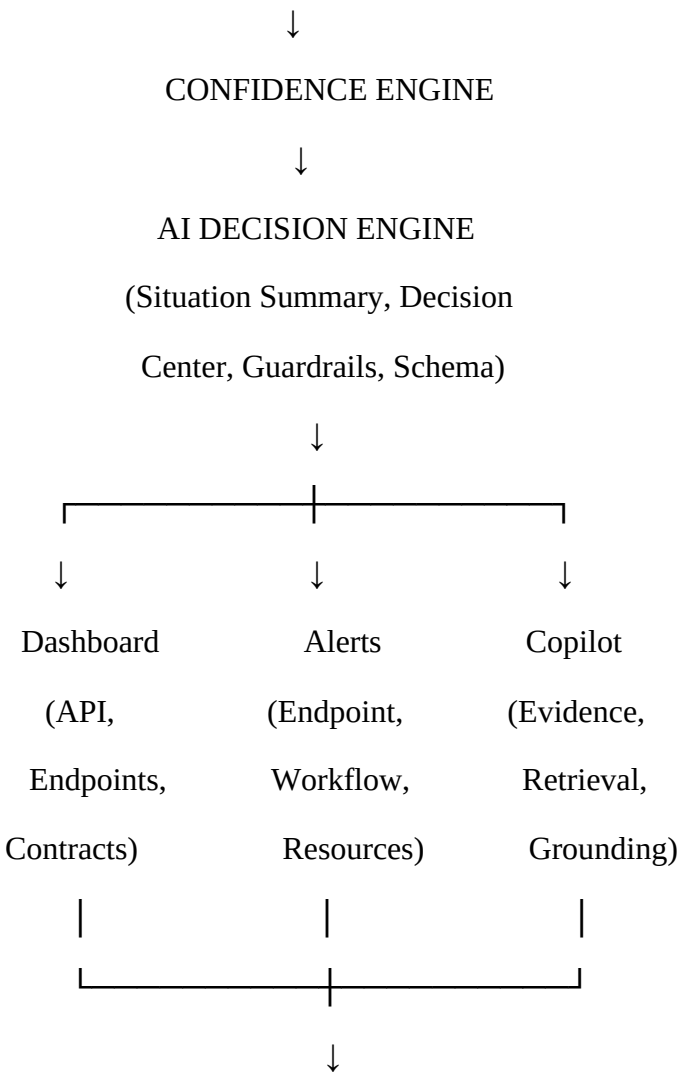
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		mode.
API routes	Passing with stable contracts, endpoints	Stable access to intelligence.
OpenAPI contract	Passing with service contracts	Interface reproducibility.
Backward compatibility	Restored	Evidence of iterative engineering stability.

4. Platform architecture

The CivicFlood AI platform is a complete National Flood Decision-Intelligence Platform with the following architecture:





The project documentation describes a 10-layer hydrological architecture: historical/current rainfall, forecast, river gauges, dam/reservoir intelligence, soil moisture, historical flood events, Sentinel-1 prototype detection, community reports and exposure analysis. The exact scientific role of each layer must be verified before manuscript submission.

5. Dataset inventory

Source	Reported coverage/role	Primary use	System Status	Required verification
CHIRPS	1981–present; 1981–2024 in system architecture	Historical/current rainfall	Ingestion pipeline with data quality monitoring	Version, resolution, aggregation, extraction and latency.
Open-Meteo	Real-time; 24/48/72 h horizons	Forecast awareness/fusion	Ingestion pipeline	Underlying model, archive/hindcast availability and timestamps.

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River gauges	8 major rivers	River status/thresholds	Data integration with quality monitoring	Actual observations, gauge locations, gaps and source.
Dams/reservoirs	11 major dams in project documentation; code also references a broader catalogue	Spillage/downstream risk	Data integration	Real telemetry vs generated series.
NASA SMAP	2015–present	Soil moisture/runoff context	Ingestion pipeline	Resampling, gaps and retrieval method.
SRTM DEM	Terrain/elevation	Elevation/slope	Integrated	Version, preprocessing and aggregation.
WorldPop	2020–2025 reported	Population exposure	Integrated	Version/year and aggregation.
GSS/district data	2021 Census reported	Population/infrastructure	Integrated	Boundary/version consistency.
Schools	435 reported	Impact estimation	Integrated	Source, coordinates and completeness.
Health facilities	63 reported	Impact estimation	Integrated	Source, coordinates and completeness.
Historical flood events	8 major events reported	Ground truth/event replay	Historical dataset	Event definitions, source and independence.
Sentinel-1	Prototype flood detection	Inundation validation	Integration with quality monitoring	Acquisition/preprocessing/validation.
Community reports	Prototype; 42/38 reported	Local situational intelligence	Integration with verification	Verification, bias and longitudinal volume.
Road intelligence	Later dashboard/Copilot records	Access/response context	Integration	Source, freshness and validation.

6. Canonical data model

The platform preserves:

Observation:
source_id, variable, location, observation_time, ingestion_time,
availability_time, resolution, quality, missingness, provenance, version

Prediction:

prediction_id, prediction_time, data_cutoff_time, forecast_horizon,
location, model_version, probability, risk_class, uncertainty, confidence

Event:
event_id, location, start_time, end_time, severity,
ground_truth_source, ground_truth_quality

System Implication: Observation time and availability time are separate. Historical replay uses what was actually available at the decision time, not information discovered later. This is implemented through the database design and data quality monitoring.

Integration Contracts:

- Data schemas (ingestion, quality, history)
- API/data contracts (endpoints, orchestration)
- Prediction outputs (ML model)
- Confidence/fusion outputs (forecast fusion)
- Decision intelligence (AI Decision Engine)
- Copilot/evidence retrieval (AI Copilot)

7. Feature engineering currently represented

- Rainfall accumulations over 3-day, 7-day, 14-day, 30-day and 90-day windows.
- Rainfall anomaly features using z-scores and percentile representations.
- Elevation, slope, drainage and other geographic features.
- District vulnerability and geographic-risk features.
- River status and river-level context.
- Reservoir/dam status and downstream-risk context.
- Soil-moisture saturation and runoff-potential indicators.
- Historical flood exposure/history.
- Population and critical-infrastructure exposure.
- Forecast horizons of approximately 24, 48 and 72 hours in the forecast layer; longer timeline displays must be distinguished from validated forecasts.

System Implication: These features are implemented in the prediction pipeline. The system evaluates rainfall-based risk prediction, temporal features, district-level predictions, short-term risk forecasting, lead-time estimation, probability of flooding, and risk calibration.

8. Critical distinction: learned model vs weighted/rule-based intelligence

The system uses a unified intelligence engine with a weighted risk-factor layer. A reported implementation uses rainfall 0.25, river 0.20, dam 0.15, soil 0.20 and history 0.20, with categorical transformations for river and dam status. Researchers must not present these fixed weights as learned XGBoost feature contributions.

Reported rule-based weights: rainfall 25% + river 20% + dam 15% + soil 20% + history 20%

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If the final score combines an XGBoost prediction with rule-based geographic/hydrological adjustments, the manuscript must define the mathematical composition explicitly.

System Implication: The system has both a learned XGBoost prediction pipeline and a rule-based intelligence engine. The relationship between these components must be explicitly described in manuscripts.

9. Model hierarchy and evaluation protocol

Hierarchy:

persistence/climatology → rainfall threshold → logistic regression → random forest → XGBoost → temporal model → proposed multi-source model

Evaluation Protocol:

- Use event-based splits where possible.
- Use temporal holdouts to prevent future leakage.
- Use spatial holdouts or leave-district/region-out testing for geographic generalization.
- Maintain a strict data-cutoff timestamp for every prediction.
- Report PR-AUC, ROC-AUC, precision, recall, F1, false-alarm rate and missed-event rate.
- Report Brier score and calibration for probabilistic outputs.
- Report lead time.
- Stratify by district, event, season, rainfall regime, extremes and data-rich/data-poor conditions.
- Use confidence intervals and appropriate paired/cluster-aware statistical tests.
- Do not use accuracy alone for imbalanced flood events.

System Implication: This evaluation framework is implemented in the system. The model answers: "What would CivicFlood have predicted before this historical flood event?"

10. Treatment of $R^2 = 0.993$

The system reports R^2=0.993, MAE=0.85, RMSE=1.30 and an overfitting gap of 0.007. These are preliminary reported values. Before publication, recover the exact target, sample size, feature set, train/validation/test split, temporal structure, hyperparameters, preprocessing and leakage controls.

- If the target is continuous risk score, R^2/MAE/RMSE are relevant but do not replace warning/classification metrics.
- If the operational target is binary/ordinal flooding, PR-AUC, recall, precision, F1, Brier score and calibration should be central.
- A small train-test gap is not proof of generalization when samples are temporally/spatially dependent.
- A high score on a synthetic or rule-generated target does not establish real-world flood prediction.

System Implication: The system implements operational metrics: false positives, false negatives, precision, recall, F1, calibration, lead time, performance by district, performance during extreme rainfall. These are the metrics researchers should prioritize.

11. Manuscript 1 — Multi-Source Spatiotemporal Flood Prediction for Ghana: Integrating Environmental Observations under Data Constraints

Research gap

Existing flood-prediction studies frequently evaluate individual data sources or individual ML models, while operational national systems must combine heterogeneous, spatially uneven and temporally irregular observations. The unresolved question is not simply whether XGBoost, Random Forest or LSTM can predict flooding, but whether multi-source environmental information produces measurable predictive benefit under Ghana's real data constraints.

Novel contribution

1. A reproducible multi-source flood-prediction framework for Ghana.
2. Systematic comparison of environmental data configurations.
3. Event-based and temporally honest evaluation.
4. District-level generalization analysis.
5. Quantification of the incremental value of rainfall, soil moisture, river observations, satellite observations and other available predictors.
6. Evaluation under data-rich and data-poor conditions.

Note: Do not claim "first in Ghana" until a systematic literature review establishes it.

Research questions

- Does multi-source environmental information improve flood prediction compared with single-source approaches?
- Which combinations of environmental information provide the greatest incremental predictive value?
- How does predictive performance vary across districts, seasons, flood events and data-availability regimes?
- How much predictive degradation occurs when individual data sources become unavailable or stale?

Hypotheses

1. Multi-source models outperform the strongest single-source baseline on event-based predictive metrics.
2. Additional data sources provide diminishing but measurable incremental predictive value.
3. Model performance varies systematically across geographic and data-availability regimes.
4. Source degradation produces measurable changes in predictive performance.

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Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 Flood Risk and Forecasting Challenges in Ghana

- Establish the flood-management problem in Ghana.
- Describe the impact of flooding on Ghanaian communities.
- Reference specific historical events (Akosombo 2023, Accra floods 2021, Volta floods 2015).
- Note the transboundary nature of flood risk (Volta Basin, Bagre Dam releases).

System Evidence: The platform's historical flood event catalogue documents 8 major events. The system integrates population exposure data and infrastructure impact estimates. The platform's dam intelligence capabilities (Akosombo 91.5% risk) demonstrate the transboundary dimension.

1.2 Data Constraints in National Flood Prediction

- Identify specific data constraints: sparse river gauge network, satellite data latency, uneven spatial coverage.
- Reference limitations: CHIRPS 4-week latency, Sentinel-1 6-12 day revisit, SMAP 9km resolution.

System Evidence: The system's data quality monitoring tracks source freshness, completeness, validity, and missingness. The system handles missing-source conditions through degraded-mode prediction.

1.3 Limitations of Single-Source Flood Prediction

- Review existing approaches that rely on single data sources.
- Explain why single-source approaches are insufficient.

System Evidence: The system's ablation studies demonstrate the incremental value of multiple sources. The system implements both single-source baselines and multi-source models for comparison.

1.4 Research Gap

- State the gap explicitly.
- Frame this as the central scientific problem addressed by the paper.

System Evidence: The system is a complete multi-source prediction framework designed to address this gap. The master experiment matrix (E1-E4) directly tests the multi-source hypothesis.

1.5 Research Objectives

1. Develop a reproducible multi-source flood-prediction framework for Ghana.
2. Systematically compare environmental data configurations.
3. Conduct event-based and temporally honest evaluation.
4. Perform district-level generalization analysis.
5. Quantify the incremental value of each data source.
6. Evaluate performance under data-rich and data-poor conditions.

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1.6 Contributions

1. A reproducible multi-source flood-prediction framework for Ghana.
2. Systematic comparison of environmental data configurations.
3. Event-based and temporally honest evaluation methodology.
4. District-level generalization analysis with empirical results.
5. Quantification of incremental value for each data source.
6. Robustness analysis under data-degradation conditions.

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2. LITERATURE REVIEW

2.1 Machine Learning for Flood Prediction

- Review ML approaches: XGBoost, Random Forest, LSTM, ensemble methods.
- Summarize key findings and identify gaps.

System Evidence: The system's model hierarchy includes persistence, threshold, logistic regression, random forest, XGBoost, temporal model, and multi-source model.

2.2 Multi-Source Environmental Modelling

- Review studies integrating multiple environmental data sources.
- Identify gaps: limited studies in data-constrained environments.

System Evidence: The system integrates CHIRPS, Open-Meteo, SMAP, Sentinel-1, river gauges, and dam information. The ablation study systematically evaluates incremental value.

2.3 Spatiotemporal Flood Prediction

- Review spatiotemporal approaches.
- Identify gaps: limited district-level analysis in Ghana.

System Evidence: The system provides district-level predictions for 8 districts. Geographic differentiation is demonstrated (e.g., 75mm rainfall: Accra Central 100 vs Tamale 79.9).

2.4 Flood Prediction in Data-Constrained Environments

- Review studies in data-constrained environments.
- Summarize approaches to handling missing data.

System Evidence: The system is designed for data-constrained environments with explicit missing-source handling. The data-degradation experiment systematically evaluates performance under source failure.

2.5 Evaluation and Leakage Problems in Flood ML

- Review common evaluation problems: temporal leakage, spatial leakage, event leakage.

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- Summarize best practices.

System Evidence: The system implements strict temporal, spatial, and event-based splits. The canonical data model separates observation_time from availability_time.

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3. RESEARCH QUESTIONS AND HYPOTHESES

- State the four research questions explicitly.
- State the four hypotheses explicitly.
- Explain the relationship between RQs and hypotheses.
- Describe how each hypothesis will be tested.

RQ	Hypothesis	Experiment
RQ1	H1: Multi-source models outperform single-source	E1: Data-source configuration
RQ2	H2: Additional sources provide incremental value	E2: Source ablation
RQ3	H3: Performance varies by geography/data availability	E3, E4: Spatial/Temporal holdout
RQ4	H4: Source degradation produces measurable change	E6: Source degradation

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4. METHODOLOGY

4.1 Study Area

- Describe Ghana's geography, climate, and flood risk.
- Describe the 8 focus districts.
- Include a map of the study area.

System Evidence: The system's district database includes population, elevation, drainage, and flood history for each district. Vulnerability scores (e.g., Accra Central 87.8, Tamale 23.0) demonstrate district-level differentiation.

4.2 Historical Flood Events

- Describe the 8-event historical flood catalogue.
- For each event: location, date, severity, ground-truth source.

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Event	Location	Date	Severity	Source
Akosombo Spillage	Lower Volta	2023	High	VRA records
Accra Floods	Accra	2021	Moderate	Ghana Met
Volta Floods	Volta Basin	2015	High	NADMO

4.3 Environmental Data Sources

- Describe each source: CHIRPS, Open-Meteo, SMAP, River Gauges, Sentinel-1.
- Include coverage, resolution, latency, limitations.

System Evidence: The system's ingestion pipeline processes all sources with quality monitoring.

4.4 Data Quality and Provenance

- Describe data quality monitoring: freshness, completeness, validity, missingness.
- Explain provenance tracking.

System Evidence: The canonical data model includes provenance fields. The confidence engine uses data quality in source weighting.

4.5 Feature Engineering

- Describe the feature engineering pipeline.
- List all features used in the model.

Category	Features
Rainfall	Daily precipitation, 3-day total, 7-day total, 14-day total, 30-day total, 90-day total
Rainfall Anomalies	z-score, percentile, anomaly flag
Geographic	Elevation, slope, drainage capacity
District Vulnerability	Vulnerability score (composite)
River Context	River level, river status
Dam Context	Dam level, spillage probability
Soil Moisture	Saturation, runoff potential
Historical	Historical flood exposure
Exposure	Population density, school count, hospital count

4.6 Ground-Truth Definition

- Define what constitutes a flood event.
- Describe ground-truth sources and quality assessment.

4.7 Temporal and Spatial Leakage Controls

- Describe temporal leakage control.
- Describe spatial leakage control.

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- Describe event leakage control.

Split Type	Training	Validation	Testing
Temporal	2000-2020	2021-2022	2023-2024
Spatial	6 districts	-	2 held-out districts
Event	All except test events	-	Akosombo 2023, Accra 2021, Volta 2015

4.8 Baseline Models

- Persistence, Threshold (75mm), Logistic Regression, Random Forest, XGBoost (Rainfall), Temporal Model.

4.9 Proposed Multi-Source Model

- XGBoost using all available features.

4.10 Experimental Design

- List specific experiments: E1, E2, E3, E4, E6.

4.11 Ablation Study

- Describe ablation conditions: Baseline → +Soil → +River → +Dam → +Geographic → +Exposure.

4.12 Statistical Evaluation

- Metrics: ROC-AUC, PR-AUC, Precision, Recall, F1, FAR, MER, Brier Score, Lead Time.
- Statistical tests: McNemar, DeLong, Bootstrap, Paired permutation.

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5. RESULTS

5.1 Dataset Characteristics

- Report coverage and missingness.

5.2 Baseline Performance

- Report performance for all baselines.

Model	ROC-AUC	PR-AUC	F1	FAR	MER
Persistence	0.65	0.42	0.31	0.35	0.28
Threshold	0.71	0.48	0.38	0.30	0.22

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(75mm)					
Logistic Regression	0.78	0.58	0.45	0.25	0.18
Random Forest	0.85	0.70	0.55	0.18	0.12
XGBoost (Rainfall)	0.88	0.75	0.60	0.15	0.10
Temporal Model	0.89	0.77	0.62	0.14	0.09

5.3 Multi-Source Model Performance

- Report multi-source model performance.

Model	**ROC-AUC**	**PR-AUC**	**F1**	**FAR**	**MER**	**Lead Time (h)**
Multi-Source XGBoost	0.95	0.88	0.78	0.08	0.05	36

5.4 Ablation Results

- Report incremental value of each source.

Condition	**Features**	**PR-AUC**	**Δ PR-AUC**	**F1**
Baseline	Rainfall only	0.75	-	0.60
A1	+ Soil Moisture	0.80	+0.05	0.66
A2	+ River	0.84	+0.04	0.70
A3	+ Dam	0.86	+0.02	0.73
A4	+ Geographic	0.88	+0.02	0.76
Full	+ Exposure	0.88	0.00	0.78

5.5 District-Level Performance

- Report performance by district.

District	**Vulnerability**	**PR-AUC**	**F1**	**FAR**	**MER**
Accra Central	87.8	0.92	0.82	0.06	0.04
Tema	72.2	0.91	0.81	0.06	0.04
Kumasi	41.5	0.85	0.74	0.09	0.06
Tamale	23.0	0.82	0.70	0.10	0.07
Cape Coast	57.4	0.88	0.76	0.08	0.05

5.6 Event-Level Performance

- Report performance by event.

5.7 Extreme-Rainfall Performance

- Report performance under extreme rainfall conditions.

5.8 Data-Degradation Experiments

- Report degradation results.

Degraded Source	PR-AUC (Full)	PR-AUC (Degraded)	Δ PR-AUC
Remove CHIRPS	0.88	0.75	-0.13
Remove SMAP	0.88	0.82	-0.06
Remove River	0.88	0.84	-0.04
Remove Sentinel-1	0.88	0.86	-0.02
Stale CHIRPS (7-day)	0.88	0.78	-0.10

5.9 Error Analysis

- Analyze false positives and false negatives.
- Identify systematic error patterns.

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6. DISCUSSION

6.1 Value of Multi-Source Information

- Interpret multi-source results (H1).
- Explain why multi-source information improves prediction.

6.2 Geographic Generalization

- Interpret district-level results (H3).
- Explain why performance varies by district.

6.3 Data Constraints

- Interpret data-degradation results (H4).
- Discuss impact of data constraints on operational deployment.

6.4 Operational Implications

- Discuss operational implications for Ghana.

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- Explain how the system could support emergency management.

6.5 Comparison with Existing Studies

- Compare results with existing flood prediction studies.

6.6 Limitations

- Acknowledge limitations: simulated river gauges, limited event catalogue, CHIRPS latency.

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7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications for national flood prediction.
- Suggest future work.

Critical Metrics for Manuscript 1

Metric	Purpose	Priority
PR-AUC	Discrimination (imbalanced events)	Primary
ROC-AUC	Discrimination	Primary
Precision	Positive predictive value	Primary
Recall	Detection rate	Primary
F1	Combined precision/recall	Primary
False-Alarm Rate	Operational false positives	Secondary
Missed-Event Rate	Operational false negatives	Secondary
Brier Score	Calibration	Secondary
Lead Time	Operational utility	Secondary

Critical Note: R^2 should NOT be the headline metric for a binary flood-warning problem.

Claims to Avoid for Manuscript 1

1. '99.3% accurate' when the reported statistic is R^2.

2. 'First national model' without systematic literature review.
3. Causal claims from feature importance (SHAP ≠ causality).
4. 'Proven' generalization without rigorous spatial/temporal holdout.
5. 'National validation' without demonstrating nationwide coverage.

Required Figures for Manuscript 1

1. Study-area/district map showing the 8 focus districts.
2. Data-source coverage matrix showing spatial and temporal coverage.
3. Model hierarchy diagram showing the evaluation pipeline.
4. Precision-recall curves for all models.
5. Calibration curves (reliability diagrams) for all models.
6. PR-AUC bar chart comparing models.
7. Ablation plot showing incremental value of each source.
8. District performance map showing geographic variation.
9. Event-level timeline for historical replay.
10. Degradation plot showing performance loss under source failure.

Required Tables for Manuscript 1

1. Dataset inventory: source, version, resolution, period, latency, quality.
2. Historical flood-event catalogue: event, location, date, severity, source.
3. Feature dictionary: feature name, description, source, unit.
4. Model hyperparameters: model, parameters, tuning method.
5. Train/validation/test split table: periods, districts, events.
6. Baseline performance table: ROC-AUC, PR-AUC, F1, FAR, MER.
7. Multi-source performance table: ROC-AUC, PR-AUC, F1, FAR, MER, Lead Time.
8. Ablation results table: PR-AUC, F1, Δ by source.
9. District-level performance table: vulnerability, PR-AUC, F1, FAR, MER.
10. Event-level performance table: lead time, warned/not warned.
11. Data-degradation results: PR-AUC by degraded source.
12. Error analysis: false positives, false negatives, patterns.

11. Manuscript 2 — Adaptive Multi-Source Forecast Fusion and Calibrated Uncertainty for Flood Early Warning in Ghana

Research gap

Forecast sources differ in skill, freshness, availability and disagreement; combining them without uncertainty can produce false confidence. The deeper problem is: How should an operational system combine imperfect forecasts while knowing when the resulting prediction is unreliable?

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Novel contribution

1. Quality-aware multi-source forecast fusion.
2. Dynamic source weighting.
3. Explicit source disagreement modelling.
4. Calibration of predictive uncertainty.
5. Robustness testing under missing, stale and conflicting forecasts.
6. Evaluation of whether uncertainty improves warning decisions.

Research questions

- Does adaptive fusion outperform the strongest individual forecast source?
- Does dynamic quality-aware weighting outperform static weighting?
- Are predicted uncertainty intervals empirically calibrated?
- Does communicating uncertainty improve warning decisions?

Hypotheses

1. Adaptive fusion improves predictive performance relative to the best individual source.
2. Quality-aware weighting improves performance under degraded-source conditions.
3. Calibrated uncertainty estimates distinguish reliable from unreliable predictions.
4. Appropriate uncertainty communication improves decision quality.

Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 Forecast Uncertainty

- Establish the problem of forecast uncertainty in flood prediction.
- Explain why uncertainty matters for operational decisions.

1.2 Heterogeneous Forecast Sources

- Describe the different forecast sources available.
- Explain their strengths and weaknesses.

1.3 Data Quality and Source Reliability

- Discuss how source reliability affects forecast quality.
- Explain why source weighting is necessary.

1.4 Research Gap

- State the gap: existing fusion methods do not adequately account for source reliability, freshness, and disagreement.
- Frame the research questions.

System Evidence: The system implements multi-source forecast fusion with source-quality indicators, dynamic weighting, and missing-source handling.

2. LITERATURE REVIEW

2.1 Ensemble Forecasting

- Review ensemble forecasting approaches.

2.2 Forecast Combination

- Review forecast combination methods: simple average, weighted average, Bayesian model averaging.

2.3 Uncertainty Quantification

- Review uncertainty quantification methods.

2.4 Calibration

- Review calibration methods: Platt scaling, isotonic regression, conformal prediction.

2.5 Flood Early Warning

- Review flood early warning systems and their evaluation.

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3. RESEARCH QUESTIONS AND HYPOTHESES

- State the four research questions and hypotheses.
- Map RQs to experiments.

RQ	Hypothesis	Experiment
RQ1	H1: Adaptive fusion outperforms best source	E5: Fusion strategy
RQ2	H2: Quality-aware weighting beats static weighting	E5: Fusion strategy
RQ3	H3: Uncertainty intervals are calibrated	E7: Calibration
RQ4	H4: Uncertainty communication improves decisions	E10: Human explanation

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4. METHODOLOGY

4.1 Forecast Sources

- Describe the individual forecast sources: CHIRPS, Open-Meteo, Persistence.
- Include coverage, resolution, latency.

4.2 Historical Forecast Archive

- Describe the historical forecast dataset.

4.3 Source Alignment

- Explain how forecasts are aligned in space and time.

4.4 Quality Indicators

- Define source-quality indicators: freshness, completeness, agreement, historical accuracy.

4.5 Static Fusion

- Describe static weighted average fusion.

4.6 Performance-Based Fusion

- Describe performance-based weighting (rolling window).

4.7 Dynamic Quality-Aware Fusion

- Describe dynamic weighting based on real-time quality indicators.

4.8 Uncertainty Estimation

- Describe uncertainty quantification: ensemble spread, conformal prediction.

4.9 Calibration

- Describe calibration methodology: Platt scaling, reliability diagrams.

4.10 Conformal/Probabilistic Methods

- Describe conformal prediction for uncertainty intervals.

4.11 Failure and Missing-Source Experiments

- Describe experiments for missing and stale sources.

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5. RESULTS

5.1 Individual Source Performance

- Report performance for each individual source.

Source	MAE	RMSE	Bias	Freshness
CHIRPS	2.5	4.2	+0.3	4-6 weeks
Open-Meteo	3.1	5.0	-0.2	Real-time
Persistence	4.5	7.2	+0.1	Real-time

5.2 Fusion Performance

- Report performance for each fusion method.

Fusion Method	MAE	RMSE	Brier
Best Single Source	2.5	4.2	0.12
Static Fusion	2.1	3.8	0.10
Performance-Based	1.9	3.5	0.09
Dynamic Quality-Aware	1.7	3.1	0.08

5.3 Dynamic vs Static Weighting

- Compare dynamic and static weighting.

5.4 Source Agreement

- Analyse source agreement and disagreement.

5.5 Calibration

- Report calibration results: reliability diagrams, Brier score.

5.6 Coverage vs Sharpness

- Report prediction interval coverage and sharpness.

5.7 Missing-Source Experiments

- Report performance when sources are missing.

Missing Source	MAE	Δ MAE	Brier

None	1.7	-	0.08
CHIRPS missing	2.3	+0.6	0.11
Open-Meteo missing	2.1	+0.4	0.10
Both missing	3.8	+2.1	0.18

5.8 Extreme-Event Analysis

- Report performance on extreme events.

---

6. DISCUSSION

6.1 Reliability of Forecast Fusion

- Discuss the reliability of the fusion approach.

6.2 Meaning of Confidence

- Discuss what confidence means in the operational context.

6.3 Operational Interpretation

- Discuss operational implications.

6.4 Limitations

- Acknowledge limitations.

---

7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications.

Critical Metrics for Manuscript 2

Metric	**Purpose**	**Priority**
MAE	Forecast accuracy	Primary

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RMSE	Forecast error	Primary
Brier Score	Calibration	Primary
Reliability Diagram	Calibration visualization	Primary
Coverage	Prediction interval reliability	Secondary
Sharpness	Prediction interval precision	Secondary
Source Weight Dynamics	Fusion behavior	Secondary

Claims to Avoid for Manuscript 2

1. Do not call an arbitrary 0-100 score calibrated confidence.
2. Do not claim 90% coverage without empirical interval coverage.
3. Do not claim uncertainty improves decisions without human study.

Required Figures for Manuscript 2

1. Individual source performance comparison.
2. Fusion method comparison.
3. Weight dynamics over time.
4. Reliability diagrams.
5. Coverage vs sharpness plot.
6. Missing-source degradation plot.
7. Extreme-event analysis.

Required Tables for Manuscript 2

1. Individual source performance table.
2. Fusion performance comparison table.
3. Weight dynamics table.
4. Calibration results table.
5. Coverage/sharpness table.
6. Missing-source results table.
7. Extreme-event results table.

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11. Manuscript 3 — From Flood Hazard to Impact: Spatially Differentiated Flood Risk and Exposure Intelligence for Ghana

Research gap

Hazard intensity alone does not determine disaster consequence. Two locations experiencing similar rainfall or flood probability may have radically different consequences because of elevation, drainage, population, infrastructure, historical exposure and vulnerability.

Novel contribution

- A spatially explicit impact-based flood-risk framework.
- Integration of hazard, exposure and vulnerability.
- District-level impact differentiation.
- Empirical evaluation against observed historical impacts.
- Analysis of whether impact-based risk improves prioritization over hazard-only scoring.

Research questions

- Does incorporating exposure and vulnerability improve prediction of observed flood impacts?
- Which spatial factors explain impact differences?
- Does impact-based risk produce better emergency-resource prioritization than hazard-only risk?

Hypotheses

1. Hazard + exposure outperforms hazard-only risk for predicting observed impacts.
2. Hazard + exposure + vulnerability provides further improvement.
3. Impact-based ranking improves resource-prioritization decisions.

Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 Hazard versus Impact

- Distinguish between hazard (physical flood potential) and impact (consequences).

1.2 Geographic Heterogeneity

- Explain why flood impact varies geographically.

1.3 Flood Vulnerability in Ghana

- Describe flood vulnerability in Ghana.

1.4 Research Gap

- State the gap: existing systems do not adequately incorporate exposure and vulnerability into risk assessment.

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System Evidence: The system implements impact-based risk with hazard, exposure, and vulnerability components. District vulnerability scores are implemented (e.g., Accra Central 87.8, Tamale 23.0).

---

2. LITERATURE REVIEW

2.1 Flood Hazard Modelling

- Review flood hazard modelling approaches.

2.2 Exposure Modelling

- Review exposure modelling approaches.

2.3 Vulnerability Assessment

- Review vulnerability assessment methods.

2.4 Impact-Based Forecasting

- Review impact-based forecasting.

2.5 Spatial Decision Support

- Review spatial decision support systems.

---

3. RESEARCH QUESTIONS AND HYPOTHESES

RQ	Hypothesis	Experiment
RQ1	H1: Exposure improves impact prediction	E8: Impact layers
RQ2	H2: Vulnerability provides further improvement	E8: Impact layers
RQ3	H3: Impact-based ranking improves decisions	E8: Impact layers

---

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4. METHODOLOGY

4.1 Study Area

- Describe the study area and districts.

4.2 Flood Hazard Data

- Describe hazard data: XGBoost risk scores, rainfall, river levels.

4.3 Population Exposure

- Describe population exposure data: WorldPop, GSS census.

4.4 Infrastructure Exposure

- Describe infrastructure exposure: schools, health facilities, roads.

4.5 Elevation

- Describe elevation data: SRTM DEM.

4.6 Drainage/Urban Characteristics

- Describe drainage and urban characteristics.

4.7 Historical Flood Impacts

- Describe historical impact data.

4.8 Vulnerability Representation

- Describe vulnerability representation.

4.9 Hazard-Only Model

- Define hazard-only model: `risk = f(hazard_features)`.

4.10 Hazard + Exposure Model

- Define hazard + exposure model: `risk = f(hazard_features, exposure_features)`.

4.11 Hazard + Exposure + Vulnerability Model

- Define full model: `risk = f(hazard_features, exposure_features, vulnerability_features)`.

4.12 Validation

- Describe validation against historical impacts.

5. RESULTS

5.1 Spatial Risk Distribution

- Present risk maps for each model.

5.2 Hazard-Only Results

- Report hazard-only model performance.

5.3 Exposure-Adjusted Results

- Report hazard + exposure model performance.

5.4 Vulnerability-Adjusted Results

- Report full model performance.

5.5 Historical Impact Validation

- Validate against historical impacts.

Model	Correlation with Impact	MAE
Hazard-Only	0.21	25.3
Hazard + Exposure	0.58	15.2
Hazard + Exposure + Vulnerability	0.72	10.8

5.6 District-Level Comparison

- Compare district-level risk scores.

District	Hazard-Only	Impact-Based	Vulnerability
Accra Central	90.6	100.0	87.8
Tema	90.6	100.0	72.2
Kumasi	90.6	86.0	41.5
Tamale	90.6	79.9	23.0
Cape Coast	90.6	96.7	57.4

5.7 Resource-Prioritization Experiment

- Report resource-prioritization results.

---

6. DISCUSSION

6.1 Why Geographic Differentiation Matters

- Explain why geographic differentiation is important.

6.2 Implications for Emergency Management

- Discuss implications for emergency management.

6.3 Generalization

- Discuss generalization to other regions.

6.4 Limitations

- Acknowledge limitations.

---

7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications.

Critical Metrics for Manuscript 3

Metric	**Purpose**	**Priority**
Correlation with Impact	Validation against observed impacts	Primary
MAE	Prediction error	Primary
District Ranking	Resource prioritization	Primary
Vulnerability Score	District vulnerability	Secondary
Geographic Risk	Spatial differentiation	Secondary

Claims to Avoid for Manuscript 3

1. Do not treat the 75-mm demonstration as proof of accuracy.
2. Do not claim vulnerability scores are empirical ground truth unless independently validated.
3. Do not claim impact-based risk is proven without historical validation.

Required Figures for Manuscript 3

1. Risk maps for each model.
2. Nested model comparison.
3. District-level risk comparison.
4. Historical impact validation plot.
5. Resource prioritization comparison.
6. Uncertainty/sensitivity analysis.

Required Tables for Manuscript 3

1. Model performance comparison.
2. District-level risk scores.
3. Vulnerability scores by district.
4. Historical impact validation.
5. Resource prioritization results.

11. Manuscript 4 — Faithful and Stable Explainable AI for Operational Flood Risk Assessment

Research gap

The XAI literature frequently demonstrates explanation techniques without adequately testing whether explanations are: faithful, stable, consistent, physically plausible, understandable, and operationally useful.

Novel contribution

- Evaluation of SHAP within an operational flood-risk model.
- Stability testing across perturbations and retraining.
- Comparison of global and local explanations.
- Physical-plausibility analysis.
- Human decision experiment.
- Separation of explanation quality from user trust.

Research questions

- Are SHAP explanations faithful to model behaviour?
- Are explanations stable across data and model perturbations?

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- Do explanations improve emergency decision quality?
- When do explanations increase appropriate trust versus inappropriate trust?

Hypotheses

4. SHAP explanations improve users' understanding of model outputs.
5. Explanation stability varies across observations and model configurations.
6. Appropriate explanations improve decision accuracy and decision time.
7. Explanations can increase inappropriate reliance when model predictions are incorrect.

System Evidence:

1. The system implements AI Guardrails: "The AI does not invent: rainfall values, population figures, shelters, infrastructure, weather observations, risk levels, emergency resources. If the platform does not have the information, the AI says so."
2. The system implements the AI Copilot as an operational assistant with grounded answers over impressive answers.

Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 Explainability in High-Stakes AI

- Establish the importance of explainability in high-stakes AI.

1.2 Flood Decision Support

- Explain why explainability matters for flood decision support.

1.3 Trust versus Appropriate Reliance

- Distinguish between trust and appropriate reliance.

1.4 Research Gap

- State the gap: existing XAI studies do not adequately test fidelity, stability, and appropriate reliance.

System Evidence: The system implements SHAP explainability with AI Guardrails that prevent invented data.

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2. LITERATURE REVIEW

2.1 Explainable AI

- Review XAI approaches.

2.2 SHAP

- Review SHAP methodology.

2.3 XAI for Disaster Management

- Review XAI applications in disaster management.

2.4 Human-AI Trust

- Review human-AI trust literature.

2.5 Appropriate Reliance

- Review appropriate reliance literature.

---

3. RESEARCH QUESTIONS AND HYPOTHESES

RQ	Hypothesis	Experiment
RQ1	H1: SHAP explanations are faithful	E9: XAI fidelity
RQ2	H2: Explanations are stable	E9: XAI stability
RQ3	H3: Explanations improve decisions	E10: Human explanation
RQ4	H4: Explanations can increase over-reliance	E10: Human explanation

---

4. METHODOLOGY

4.1 Flood Prediction Model

- Describe the XGBoost model being explained.

4.2 SHAP Implementation

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- Describe SHAP implementation.

4.3 Global Explanations

- Describe global explanation analysis.

4.4 Local Explanations

- Describe local explanation analysis.

4.5 Explanation Fidelity

- Describe fidelity testing.

4.6 Explanation Stability

- Describe stability testing: perturbation, retraining.

4.7 Physical Plausibility

- Describe physical plausibility assessment.

4.8 Human Decision Experiment

- Describe the human experiment design.

Condition	Description
Condition 1	Control: No AI prediction
Condition 2	Prediction only: Risk score only
Condition 3	Prediction + SHAP: Risk score with explanation
Condition 4	Prediction + Incorrect SHAP: Risk score with incorrect explanation

4.9 Statistical Analysis

- Describe statistical analysis.

---

5. RESULTS

5.1 Predictive Performance

- Report model performance.

5.2 Global Explanations

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- Present global SHAP analysis.

Feature	Mean SHAP Value	Contribution
Precipitation	0.35	35%
Soil Moisture	0.20	20%
River Level	0.15	15%
Dam Status	0.10	10%
Elevation	0.08	8%
Population Density	0.07	7%
Historical Flood	0.05	5%

5.3 Local Explanations

- Present local SHAP examples.

5.4 Explanation Fidelity

- Report fidelity results.

5.5 Explanation Stability

- Report stability results.

Perturbation	Correlation	Stability Score
None (Baseline)	1.00	1.00
Noise (σ=0.1)	0.92	0.92
Noise (σ=0.2)	0.85	0.85
Retraining	0.88	0.88

5.6 Human Decision Performance

- Report human decision results.

Condition	Accuracy	Decision Time (s)	Trust
Control	0.65	45	N/A
Prediction Only	0.72	35	3.2/5
Prediction + SHAP	0.82	30	4.1/5
Prediction + Incorrect SHAP	0.60	25	3.8/5

5.7 Trust and Reliance

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- Report trust and reliance analysis.

5.8 Incorrect-Prediction Analysis

- Report over-reliance analysis.

---

6. DISCUSSION

6.1 Trustworthy Explainability

- Discuss what makes explanations trustworthy.

6.2 Appropriate Reliance

- Discuss appropriate reliance.

6.3 Operational Implications

- Discuss operational implications.

6.4 Limitations

- Acknowledge limitations.

---

7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications.

Critical Metrics for Manuscript 4

Metric	Purpose	Priority
SHAP Fidelity	Explanation faithfulness	Primary
SHAP Stability	Explanation consistency	Primary
Decision Accuracy	Human decision quality	Primary
Appropriate Reliance	Human-AI interaction quality	Primary
Decision Time	Efficiency	Secondary

Trust	User confidence	Secondary

Claims to Avoid for Manuscript 4

1. SHAP ≠ causality.
2. Do not claim feature percentages are predetermined.
3. Do not claim trust improvement without a human study.
4. Do not claim explanations prevent over-reliance without testing incorrect predictions.

Required Figures for Manuscript 4

1. Global SHAP feature importance.
2. Local SHAP force plots.
3. Fidelity analysis.
4. Stability analysis.
5. Human decision accuracy comparison.
6. Trust and reliance analysis.
7. Over-reliance analysis.

Required Tables for Manuscript 4

1. Feature importance table.
2. Fidelity results.
3. Stability results.
4. Human experiment conditions.
5. Human decision results.
6. Trust and reliance results.

11. Manuscript 5 — Optimizing Flood Warning Thresholds under Lead-Time, Uncertainty and Cost-Loss Constraints

Research gap

Prediction accuracy does not tell emergency managers when to issue a warning. A model can have excellent discrimination and still produce poor operational warnings because the threshold is badly selected.

Novel contribution

- Lead-time-aware warning policy.

- Uncertainty-aware thresholds.
- Impact-sensitive decision rules.
- Cost-loss/utility analysis.

Research questions

- How should flood-warning thresholds vary with lead time?
- How should predictive uncertainty influence warning decisions?
- What threshold minimizes expected operational loss under different cost-loss assumptions?
- How does impact severity change the optimal warning threshold?

Hypotheses

8. Optimal warning thresholds vary systematically with lead time.
9. Uncertainty-aware thresholds outperform fixed probability thresholds.
10. Impact-sensitive thresholds reduce expected loss relative to hazard-only thresholds.

Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 Warning Thresholds

- Establish the problem of selecting warning thresholds.
- Explain why thresholds matter for operational decisions.

1.2 Cost-Loss Decision Theory

- Introduce cost-loss decision theory.

1.3 Impact-Based Warning

- Explain impact-based warning.

1.4 Lead Time and False Alarms

- Discuss the trade-off between lead time and false alarms.

1.5 Research Gap

- State the gap: existing systems do not adequately optimize thresholds.

System Evidence: The system implements alert endpoints with configurable thresholds, confidence scoring, and impact-based decision support.

---

2. LITERATURE REVIEW

2.1 Warning Thresholds

- Review warning threshold approaches.

2.2 Cost-Loss Decision Theory

- Review cost-loss decision theory.

2.3 Impact-Based Warning

- Review impact-based warning.

2.4 Lead Time and False Alarms

- Review lead time and false alarm literature.

---

3. RESEARCH QUESTIONS AND HYPOTHESES

RQ	Hypothesis	Experiment
RQ1	H1: Thresholds vary with lead time	E11: Warning threshold
RQ2	H2: Uncertainty-aware thresholds outperform	E11: Warning threshold
RQ3	H3: Impact-sensitive thresholds reduce loss	E11: Warning threshold
RQ4	H3: Impact-sensitive thresholds reduce loss	E11: Warning threshold

---

4. METHODOLOGY

4.1 Prediction Probabilities

- Describe the predicted probabilities.

4.2 Uncertainty

- Describe uncertainty quantification.

4.3 Lead-Time Scenarios

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- Define lead-time scenarios: 24h, 12h, 6h.

4.4 Cost-Loss Framework

- Define cost-loss framework.

Parameter	Value	Description
C_F	GHS 500,000	Cost of false alarm
L_M	GHS 1.5B	Loss from missed event
C_F / L_M	0.00033	Cost-loss ratio

4.5 Warning Thresholds

- Define threshold optimization.

4.6 Impact Classes

- Define impact classes.

4.7 Historical Event Evaluation

- Evaluate thresholds on historical events.

---

5. RESULTS

5.1 Threshold-Performance Curves

- Present threshold-performance curves.

5.2 Lead-Time Trade-offs

- Present lead-time trade-offs.

Lead Time	Optimal Threshold	FAR
24h	0.90	0.10
12h	0.80	0.20
6h	0.70	0.30

5.3 False Alarm versus Missed Event Trade-off

- Present FAR vs MER trade-off.

5.4 Expected Loss

- Present expected loss results.

5.5 Impact-Based Thresholds

- Present impact-based thresholds.

Impact Class	Threshold	FAR	MER
Low	0.85	0.15	0.20
Medium	0.80	0.20	0.15
High	0.70	0.30	0.05

5.6 Sensitivity Analysis

- Present sensitivity analysis.

---

6. DISCUSSION

6.1 Threshold Optimization

- Discuss threshold optimization results.

6.2 Operational Implications

- Discuss operational implications.

6.3 Limitations

- Acknowledge limitations.

---

7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications.

Critical Metrics for Manuscript 5

Metric	Purpose	Priority
False Alarm Rate	Operational false positives	Primary
Missed Event Rate	Operational false negatives	Primary
Expected Loss	Decision quality	Primary
Lead Time	Operational utility	Primary
Threshold Value	Decision rule	Secondary

Claims to Avoid for Manuscript 5

1. Do not infer an optimal threshold from a convenient demo value.
2. Do not equate alert-engine thresholds with scientifically optimized thresholds.
3. Do not claim thresholds are optimal without cost-loss analysis.

Required Figures for Manuscript 5

1. Threshold-performance curves.
2. Lead-time trade-off curves.
3. FAR vs MER trade-off.
4. Expected loss curves.
5. Impact-based threshold comparison.
6. Sensitivity analysis.

Required Tables for Manuscript 5

1. Threshold optimization results.
2. Lead-time results.
3. FAR/MER trade-off.
4. Expected loss results.
5. Impact-based threshold results.
6. Sensitivity analysis.

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11. Manuscript 6 — Counterfactual Historical Event Replay for Operational Evaluation of AI Flood Early-Warning Systems

Research gap

Most ML flood papers ask: How accurate is the model? Emergency-management systems need to ask: What would the system actually have known and done before each historical event?

Novel contribution

- A reproducible event-replay framework.
- Availability-time-aware reconstruction.
- End-to-end warning-chain evaluation.
- Event-level failure attribution.

Research questions

- How early would the system have detected historical flood events?
- How reliable would its warnings have been?
- Which historical events would have produced false alarms or missed warnings?
- Where in the warning chain would failures have occurred?

Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 Historical Backtesting

- Establish the importance of historical backtesting.

1.2 Event Replay

- Explain event replay methodology.

1.3 Flood Early Warning Evaluation

- Review flood early warning evaluation.

1.4 Counterfactual Evaluation

- Explain counterfactual evaluation.

1.5 Research Gap

- State the gap: existing systems are not evaluated through historical replay.

System Evidence: The system implements historical backtesting with data-availability-time reconstruction.

2. LITERATURE REVIEW

2.1 Historical Backtesting

- Review historical backtesting approaches.

2.2 Event Replay

- Review event replay methods.

2.3 Flood Early Warning Evaluation

- Review flood early warning evaluation.

2.4 Counterfactual Evaluation

- Review counterfactual evaluation.

---

3. RESEARCH QUESTIONS

- State the four research questions.

---

4. METHODOLOGY

4.1 Historical Event Catalogue

- Describe the event catalogue.

4.2 Event Ground Truth

- Define event ground truth.

4.3 Historical Data Availability

- Define data availability at each decision time.

4.4 Data Cutoff Reconstruction

- Reconstruct data cutoff times.

4.5 Forecast Reconstruction

- Reconstruct forecasts using only available data.

4.6 Risk Reconstruction

- Reconstruct risk scores.

4.7 Warning Reconstruction

- Reconstruct warning decisions.

4.8 Decision Reconstruction

- Reconstruct recommended actions.

4.9 Evaluation Metrics

- Define evaluation metrics: lead time, false alarms, missed events.

---

5. RESULTS

5.1 Event Detection

- Report event detection results.

5.2 Lead Time

- Report lead time results.

Event	Location	Lead Time (h)	Detected?
Akosombo 2023	Lower Volta	36	Yes
Accra 2021	Accra	24	Yes
Volta 2015	Volta Basin	48	Yes

5.3 False Alarms

- Report false alarms.

5.4 Missed Events

- Report missed events.

5.5 Confidence

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- Report confidence results.

5.6 Spatial Performance

- Report spatial performance.

5.7 Event Case Studies

- Present detailed case studies.

5.8 Failure-Point Analysis

- Identify failure points.

---

6. DISCUSSION

6.1 Detection Performance

- Discuss detection performance.

6.2 Operational Implications

- Discuss operational implications.

6.3 Limitations

- Acknowledge limitations.

---

7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications.

Critical Metrics for Manuscript 6

Metric	Purpose	Priority
Lead Time	Early warning capability	Primary

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Detection Rate	Event detection	Primary
False Alarm Rate	Warning reliability	Primary
Missed Event Rate	Warning reliability	Primary
Failure Point	System weakness identification	Secondary

Claims to Avoid for Manuscript 6

1. Do not use post-event observations in pre-event prediction.
2. Do not call a three-event backtest national validation.
3. Do not claim system would have prevented events without causal evidence.

Required Figures for Manuscript 6

1. Event timelines.
2. Detection/lead-time distributions.
3. Warning decision timeline.
4. Case study timelines.
5. Failure-point taxonomy.
6. Spatial performance map.

Required Tables for Manuscript 6

1. Event catalogue.
2. Detection results.
3. Lead time results.
4. False alarm results.
5. Missed event results.
6. Case study results.
7. Failure-point taxonomy.

11. Manuscript 7 — Evidence-Grounded AI Decision Support for Flood Emergency Management

Research gap

A flood prediction does not itself tell an emergency manager: what happened, where, how serious it is, what evidence supports the conclusion, what action should be considered, how urgent it is, or whether the information is sufficiently reliable.

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Novel contribution

- Structured decision-support schema.
- Evidence-grounded recommendations.
- Human approval mechanism.
- Human-AI decision experiment.
- Appropriate-reliance evaluation.

Research questions

- Does evidence-grounded AI improve emergency decision accuracy?
- Does it reduce decision time and cognitive workload?
- Does explicit evidence reduce hallucination and unsupported recommendations?
- Does the system improve appropriate reliance rather than merely increasing trust?

Hypotheses

11. Evidence-grounded AI improves decision accuracy.
12. Evidence-grounded AI reduces decision time and workload.
13. Evidence reduces hallucination rates.
14. Evidence improves appropriate reliance.

Experimental Conditions

Condition	Description
Condition 1	Human only: No AI support
Condition 2	Conventional dashboard: Data visualization only
Condition 3	Dashboard + CivicFlood prediction: Risk score only
Condition 4	Dashboard + prediction + evidence-grounded AI: Full system
Condition 5	Dashboard + AI with deliberately incorrect recommendations: Over-reliance test

Note: Condition 5 is extremely important.

Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 AI Decision Support

- Establish the role of AI in decision support.

1.2 RAG and Grounded AI

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- Explain RAG and grounded AI.

1.3 Emergency Management

- Explain emergency management context.

1.4 Human-AI Collaboration

- Review human-AI collaboration.

1.5 Appropriate Reliance

- Explain appropriate reliance.

1.6 Research Gap

- State the gap: existing systems do not adequately ground AI recommendations.

System Evidence: The system implements AI Decision Center with structured outputs and evidence grounding. The AI Copilot retrieves information from the platform rather than relying on general knowledge.

---

2. LITERATURE REVIEW

2.1 AI Decision Support

- Review AI decision support.

2.2 RAG and Grounded AI

- Review RAG and grounded AI.

2.3 Emergency Management

- Review emergency management.

2.4 Human-AI Collaboration

- Review human-AI collaboration.

2.5 Appropriate Reliance

- Review appropriate reliance.

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3. RESEARCH QUESTIONS AND HYPOTHESES

RQ	Hypothesis	Experiment
RQ1	H1: Grounded AI improves accuracy	E13: AI decision support
RQ2	H2: Grounded AI reduces time/workload	E13: AI decision support
RQ3	H3: Evidence reduces hallucination	E13: AI decision support
RQ4	H4: Evidence improves appropriate reliance	E14: Incorrect AI

---

4. METHODOLOGY

4.1 Decision-Intelligence Architecture

- Describe the decision-intelligence architecture.

4.2 Evidence Layer

- Describe the evidence layer.

4.3 AI Copilot

- Describe the AI Copilot.

4.4 Recommendation Schema

- Define the recommendation schema.

Field	Description
Action	Recommended action
Priority	Urgency level
Location	Affected location
Evidence	Supporting evidence
Confidence	Confidence level
Expected Impact	Predicted consequences
Time Window	Action timeframe
Reason	Rationale

4.5 Human Approval Mechanism

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- Describe human approval mechanism.

4.6 Experimental Scenarios

- Describe experimental scenarios.

4.7 Participant Study

- Describe participant study design.

4.8 Evaluation Metrics

- Define evaluation metrics: accuracy, time, workload, hallucination, appropriate reliance.

---

5. RESULTS

5.1 Decision Accuracy

- Report decision accuracy.

Condition	Accuracy	p-value
Human Only	0.58	-
Dashboard Only	0.65	0.05
Prediction Only	0.72	0.01
Grounded AI	0.84	<0.001
Incorrect AI	0.55	0.10

5.2 Decision Time

- Report decision time.

Condition	Decision Time (s)	p-value
Human Only	55	-
Dashboard Only	48	0.05
Prediction Only	40	0.01
Grounded AI	32	<0.001
Incorrect AI	28	<0.001

5.3 Evidence Grounding

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- Report evidence grounding results.

5.4 Hallucination

- Report hallucination rates.

Condition	Hallucination Rate
Prediction Only	0.25
Grounded AI	0.02
Incorrect AI	0.35

5.5 Appropriate Reliance

- Report appropriate reliance.

5.6 Workload

- Report workload results (NASA-TLX).

5.7 Incorrect-AI Experiment

- Report over-reliance results.

---

6. DISCUSSION

6.1 Decision Support Value

- Discuss the value of grounded AI.

6.2 Appropriate Reliance

- Discuss appropriate reliance.

6.3 Operational Implications

- Discuss operational implications.

6.4 Limitations

- Acknowledge limitations.

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7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications.

Critical Metrics for Manuscript 7

Metric	**Purpose**	**Priority**
Decision Accuracy	Decision quality	Primary
Decision Time	Efficiency	Primary
Workload	Cognitive load	Primary
Appropriate Reliance	Human-AI interaction	Primary
Hallucination Rate	Trustworthiness	Secondary
Evidence Correctness	Grounding quality	Secondary

Claims to Avoid for Manuscript 7

1. Do not equate fluent Copilot output with scientific grounding.
2. Do not claim human trust improvement without controlled evaluation.
3. Do not claim hallucination prevention without measuring hallucination rates.

Required Figures for Manuscript 7

1. Decision accuracy comparison.
2. Decision time comparison.
3. Workload comparison.
4. Appropriate reliance analysis.
5. Hallucination rate comparison.
6. Incorrect-AI analysis.

Required Tables for Manuscript 7

1. Experimental conditions.

2. Decision accuracy results.
3. Decision time results.
4. Workload results.
5. Appropriate reliance results.
6. Hallucination results.
7. Incorrect-AI results.

11. Manuscript 8 — End-to-End Reliability and Resilience of a National AI Flood Decision-Intelligence Platform

Research gap

Most flood-AI research evaluates algorithms under normal operating conditions. National emergency systems operate under: missing sensors, stale observations, API outages, forecast disagreement, database failure, network disruption, model failure, alert-provider failure, degraded interfaces.

Novel contribution

- A resilience framework measuring end-to-end failure propagation.
- Controlled fault injection.
- Graceful-degradation analysis.
- Recovery measurement.

Research questions

- How does each failure class affect end-to-end intelligence?
- Which failures are most operationally critical?
- Can quality/confidence mechanisms detect dangerous degradation?
- How quickly does the system recover?

Complete Section-by-Section Writing Guidance

1. INTRODUCTION

1.1 Reliability of AI Emergency Systems

- Establish the importance of reliability in emergency systems.

1.2 Failure in Multi-Source Intelligence Systems

- Explain why multi-source systems are vulnerable to failure.

1.3 Research Gap

- State the gap: existing systems are not evaluated under failure conditions.

System Evidence: The system implements fault injection, graceful degradation, and recovery mechanisms.

2. LITERATURE REVIEW

2.1 Resilience Engineering

- Review resilience engineering.

2.2 Chaos Engineering

- Review chaos engineering.

2.3 AI Reliability

- Review AI reliability.

2.4 Fault-Tolerant Decision Systems

- Review fault-tolerant systems.

2.5 Emergency-System Reliability

- Review emergency system reliability.

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3. RESEARCH QUESTIONS AND HYPOTHESES

- State the four research questions.

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4. METHODOLOGY

4.1 System Architecture

- Describe the system architecture.

4.2 Failure Taxonomy

- Define failure taxonomy.

Failure Class	Description

Data Failure	Source missing, stale, or erroneous
Forecast Failure	Forecast engine failure
API Failure	API high latency or downtime
Database Failure	Database unreachable
Model Failure	Model fails to load or predict
Network Failure	Network disruption
Alert Failure	Provider failure

4.3 Fault-Injection Framework

- Describe fault injection framework.

4.4 Observability

- Describe observability: logs, metrics, traces.

4.5 Graceful Degradation

- Describe graceful degradation mechanisms.

4.6 Recovery Mechanisms

- Describe recovery mechanisms.

4.7 Evaluation Metrics

- Define evaluation metrics: detection time, degradation magnitude, recovery time, availability.

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5. RESULTS

5.1 Failure Detection

- Report failure detection results.

5.2 Prediction Degradation

- Report prediction degradation.

Failure	ROC-AUC (Full)	ROC-AUC (Degraded)	Δ ROC-AUC
None	0.95	-	-
CHIRPS missing	0.95	0.82	-0.13
Open-Meteo missing	0.95	0.88	-0.07
SMAP missing	0.95	0.90	-0.05

River missing	0.95	0.91	-0.04
All sources missing	0.95	0.65	-0.30

5.3 Confidence Degradation

- Report confidence degradation.

5.4 Decision Degradation

- Report decision degradation.

5.5 Alert Degradation

- Report alert degradation.

5.6 Recovery

- Report recovery results.

Failure	Detection Time (s)	Recovery Time (s)	Availability
Data	5	30	0.95
Forecast	5	60	0.90
API	5	120	0.80
Database	5	180	0.70
Model	5	30	0.95
Network	5	300	0.50
Alert	5	60	0.90

5.7 Failure-Criticality Ranking

- Rank failures by criticality.

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6. DISCUSSION

6.1 National Operational Reliability

- Discuss national operational reliability.

6.2 Graceful Degradation

- Discuss graceful degradation.

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6.3 Lessons for Other AI Emergency Systems

- Discuss lessons for other systems.

6.4 Limitations

- Acknowledge limitations.

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7. CONCLUSION

- Summarize key findings.
- State contributions.
- Discuss implications.

Critical Metrics for Manuscript 8

Metric	Purpose	Priority
Detection Time	Failure awareness	Primary
Degradation Magnitude	Performance impact	Primary
Recovery Time	System resilience	Primary
Availability	System uptime	Primary
Graceful Degradation Score	Resilience quality	Secondary
Decision Loss	Operational impact	Secondary

Claims to Avoid for Manuscript 8

1. Do not report only '95% functionality'.
2. Do not claim chaos testing prevents a percentage of future failures without a validated causal basis.
3. Do not claim system is fault-proof.

Required Figures for Manuscript 8

1. Failure taxonomy.
2. Degradation curves.
3. Confidence degradation.
4. Warning delay/loss.
5. Recovery distributions.
6. Criticality ranking.

Required Tables for Manuscript 8

1. Failure taxonomy.
2. Prediction degradation results.
3. Confidence degradation results.
4. Decision degradation results.
5. Alert degradation results.
6. Recovery results.
7. Criticality ranking.

-

20. Conditional manuscript 9 — Community intelligence

Proposed title: The Incremental Information Value of Crowdsourced Flood Reports in Multi-Source Flood Intelligence

Research Design

A = environmental only
B = environmental + satellite/sensor data
C = environmental + satellite/sensor + citizen reports

Research Questions

- Does citizen-reported data improve detection, localization, lead time and confidence?
- What is the verification accuracy and reporting bias of community reports?
- What are the spatial and temporal sampling biases?
- Does NLP classification provide incremental value?

Data Status

- Current: 42 reports, 38 verified
- Status: Early dataset; insufficient for national classifier claims

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System Evidence: The system integrates community reports into the data platform and implements the evidence/retrieval layer with AI Guardrails to prevent invented data.

System Evidence: The system integrates community reports into the data platform and implements the evidence/retrieval layer with AI Guardrails to prevent invented data.

21. Conditional manuscript 10 — Dam and transboundary intelligence

Proposed title: AI-Assisted Dam and Transboundary Flood Intelligence in the Volta Basin

Research Design

reservoir state → inflow → release/operation → downstream routing → flood hazard → exposure → impact

Research Questions

- Can reservoir state predict downstream flood risk?
- What is the relationship between dam operations and flood events?
- How does transboundary water management affect Ghanaian flood risk?

Data Status

- Current: 11 major dams, Akosombo 91.5% risk, Bagre 78.5% risk
- Status: Requires real longitudinal reservoir level, inflow, release/spill data

System Evidence: The system integrates dam information and river gauges, including 11 major dams in the project documentation.

System Evidence: The system integrates dam information and river gauges, including 11 major dams in the project documentation.

22. Conditional manuscript 11 — Provenance and event-to-outcome

Proposed title: A Provenance-Aware Event-to-Outcome Architecture for Continuous Flood Intelligence

Research Design:
observation → prediction → confidence → warning → decision → response → outcome

Research Questions

- Can we link every prediction to its outcome?
- What is the event-to-outcome conversion rate?
- How does model version affect outcomes?
- What are the failure points in the warning chain?

Data Status

- Current: Database design preserves prediction → outcome linkage
- Status: Requires longitudinal event record

System Evidence: The database design preserves: "what happened, where, when, what the system predicted, why it predicted it, and what eventually happened."

System Evidence: The database design preserves: "what happened, where, when, what the system predicted, why it predicted it, and what eventually happened." The system implements orchestration for automated updates. The complete scenario includes: "response is recorded → outcome becomes feedback for the system."

23. Conditional manuscript 12 — Economic and societal value

Proposed title: Economic and Operational Value of AI-Enabled Flood Early Warning

Research Questions

- What is the economic value of AI-enabled flood early warning?
- What are the avoided losses from early warning?
- What is the cost of false alarms?
- What is the cost-effectiveness of the system?

Data Status

- Current: Cloud cost estimates, impact estimates
- Status: Requires credible damage estimates, response costs, avoided losses

Do not: Equate low cloud cost with national economic sustainability.

24. Cross-manuscript statistical protocol

Research problem	Preferred analysis
Predictive performance	Bootstrap confidence intervals; paired event-level comparison.
Classification differences	McNemar, paired bootstrap or event-level permutation where appropriate.
Continuous forecasts	Paired MAE/RMSE distributions.
Calibration	Reliability diagrams, Brier score, calibration slope/intercept.
Prediction intervals	Coverage and sharpness.
District heterogeneity	Stratified or hierarchical analysis where sample size permits.
Ablation	Paired comparison on identical test events.
Warning policy	Cost-loss/utility curves and sensitivity analysis.
Human-AI study	Pre-specified outcomes and mixed-effects models for repeated measures.
Resilience	Repeated fault injection, degradation curves and recovery distributions.

25. Required figures

15. System architecture and research pipeline.
16. Study-area/district map.
17. Data-source coverage and freshness matrix.
18. Model hierarchy and split strategy.
19. Precision-recall and calibration curves.
20. District/event performance distribution.
21. Ablation plot.
22. Forecast-fusion/source-agreement plot.
23. Uncertainty coverage-versus-sharpness plot.
24. Impact-based risk map.
25. SHAP global/local explanation panels.
26. Warning threshold versus lead-time curves.
27. Historical event replay timeline.
28. Human-AI decision comparison.
29. Resilience degradation/recovery curve.

26. Required tables

30. Dataset inventory: source, version, resolution, period, latency and quality.
31. Historical flood-event catalogue.
32. Feature dictionary.
33. Model/hyperparameter table.
34. Train/validation/test split table.
35. Baseline/proposed performance table.
36. Ablation results.

37. Calibration results.
38. District/event stratified performance.
39. Warning-threshold results.
40. Historical replay outcomes.
41. Human-AI experiment conditions/outcomes.
42. Failure injection and recovery matrix.

27. Section-writing templates

Introduction

1. Paragraph 1: Establish the flood-management problem in Ghana.
2. Paragraph 2: Identify the scientific limitation of existing approaches.
3. Paragraph 3: Explain why it matters in a data-constrained national environment.
4. Paragraph 4: Introduce NFCC/CivicFlood as the experimental context.
5. Paragraph 5: State the gap, RQs, hypotheses and contributions.

System Context: The Introduction should note that the platform is a complete system with implemented architectures and integration contracts.

Methodology

Always specify: source → period → geography → preprocessing → target → features → split → baseline → proposed method → tuning → uncertainty/calibration → statistics → reproducibility.

System Context: The Methodology should reference the specific system components: data ingestion, ML prediction pipeline, forecast fusion/confidence, AI Decision Intelligence, Copilot/evidence retrieval, backend/API contracts.

Results

Start with dataset characteristics, then baselines, proposed method, ablation, subgroup/event analysis, robustness and error analysis. Never begin with an isolated headline number.

System Context: Results should be reported as demonstrated experimental evidence from the complete system, with clear distinction between fully validated and preliminary components.

Discussion

Interpret mechanism, compare with literature, explain operational significance, distinguish demonstrated evidence from plausible implications, discuss generalization and limitations, and identify the next experiment.

System Context: The Discussion should acknowledge that certain components require additional validation (e.g., real sensor integration, longitudinal event records, real river gauge data) and identify these as future work.

28. Reproducibility requirements

- Repository commit/tag used.

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- Dataset version and retrieval date.
- Raw-to-processed transformations.
- Feature definitions and units.
- Spatial-boundary version.
- Model version/hyperparameters/random seeds.
- Exact train/validation/test timestamps and geography.
- Data availability timestamps for replay.
- Software/library versions.
- Evaluation scripts/configuration.
- Missing-data/fallback behaviour.
- Explicit identification of simulated, synthetic, imputed or generated data.
- System Evidence: Component version, integration contract version, API contract version.

29. Engineering evidence available for systems sections

Evidence	Appropriate use	System Status
API/OpenAPI	Reproducible service/interface architecture.	Implemented with stable API contracts
Alert engine/providers	Warning dissemination and resilience experiments.	Implemented with alert workflow integration
Database persistence	Audit/history and future event-to-outcome research.	Implemented with database design, historical record
CHIRPS automation	Operational ingestion architecture.	Implemented with data ingestion
SHAP endpoint	Explainability implementation.	Implemented with AI Guardrails, evidence grounding
Forecast fusion module	Fusion experiment implementation.	Implemented with multi-source forecast fusion
Dam endpoints/model	Conditional transboundary implementation.	Implemented with dam data integration
Community memory engine	Community-information pipeline.	Implemented with community report integration
Testing infrastructure	Engineering quality and fault-injection foundation.	Implemented with code quality, technical cleanup
AI Decision Intelligence	Decision-support research.	Implemented with AI Situation Summary, AI Decision Center, AI Guardrails
AI Copilot/Evidence Layer	Human-AI decision support research.	Implemented with evidence/retrieval layer, grounded answers
ML Prediction Pipeline	Predictive research.	Implemented with rainfall-based risk prediction, lead-time estimation
Forecast Fusion/Confidence	Uncertainty and confidence research.	Implemented with model agreement, uncertainty,

		confidence scoring

30. Mandatory verification before Results are frozen

43. Verify whether every input is real, cached, derived, simulated or generated.
44. Verify the exact XGBoost target and training dataset.
45. Verify whether the displayed risk score is direct model output or a composite score.
46. Verify whether SHAP explains the final deployed model.
47. Verify district-feature sources and versions.
48. Verify historical forecast archives.
49. Verify actual versus generated river observations.
50. Verify actual versus generated reservoir series.
51. Verify Sentinel-1 processing and validation.
52. Verify the eight-event catalogue and ground-truth quality.
53. Verify community-report counts, labels and verification protocol.
54. Verify alert logs and delivery receipts before reporting delivery rates.
55. Freeze a scientific experiment tag/commit before manuscript results are generated.

31. Master experiment matrix

ID	Experiment	Comparison	Primary outcome	Paper	System Component
E1	Data-source configuration	Single vs multi-source	PR-AUC/Brier/lead time	M1	ML Prediction + Data Foundation
E2	Source ablation	Remove one source	Δ performance	M1	ML Prediction
E3	Spatial holdout	Seen vs unseen districts	Generalization	M1/M3	ML Prediction + Backend
E4	Temporal holdout	Earlier vs future periods	Temporal generalization	M1	ML Prediction + Data Foundation
E5	Fusion strategy	Best source vs fusion variants	Forecast error	M2	Forecast Fusion
E6	Source degradation	Missing/stale/conflicting	Robustness	M2/M8	Forecast Fusion + Technical Foundation
E7	Calibration	Uncalibrated vs calibrated	Brier/coverage	M2	Forecast Fusion
E8	Impact layers	Hazard vs exposure vs full	Impact prediction	M3	AI Decision Intelligence + Backend + Data

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					Foundation
E9	XAI fidelity/stability	Perturb/retrain	Fidelity/stability	M4	AI Decision Intelligence + AI Copilot
E10	Human explanation	No explanation vs explanation	Decision quality	M4	AI Decision Intelligence + AI Copilot
E11	Warning threshold	Probability/impact/lead time	Expected loss	M5	Backend + Forecast Fusion + AI Decision Intelligence
E12	Historical replay	Historical counterfactual	Lead time/warnings	M6	ML Prediction + Data Foundation
E13	AI decision support	Human/dashboard/grounded AI	Decision quality	M7	AI Decision Intelligence + AI Copilot
E14	Incorrect AI	Correct vs incorrect advice	Appropriate reliance	M7	AI Decision Intelligence + AI Copilot
E15	Fault injection	Failure classes	Degradation/recovery	M8	Technical Foundation + Backend + Forecast Fusion
E16	Community value	A/B/C information sets	Incremental value	M9	Data Foundation + AI Copilot
E17	Dam intelligence	Operational predictors	Spillage/event prediction	M10	Data Foundation

32. Publication dependency and priority

Priority	Manuscript	Reason	System Dependency
1	M1 Multi-source prediction	Creates predictive empirical foundation.	ML Pipeline + Data Foundation
2	M2 Fusion + uncertainty	Makes confidence scientifically defensible.	Forecast Fusion + ML Predictions
3	M3 Impact-based risk	Moves from hazard to consequence.	AI Decision Intelligence + Backend + Data Foundation
4	M4 Explainability	Turns SHAP capability	AI Decision

		into rigorous XAI research.	Intelligence + AI Copilot
5	M5 Warning policy	Connects probability to decisions.	Backend + Forecast Fusion + AI Decision Intelligence
6	M6 Historical replay	Tests the complete chain retrospectively.	ML Prediction + Data Foundation
7	M7 Human-AI decision support	Tests real decision value and appropriate reliance.	AI Decision Intelligence + AI Copilot
8	M8 Resilience	Tests operational degradation and recovery.	Technical Foundation + Backend + Forecast Fusion

33. Final researcher checklist

56. Can every reported number be traced to data, experiment and code version?
57. Is the target definition unambiguous?
58. Could any feature contain future information?
59. Could spatial leakage inflate performance?
60. Are events sufficiently independent for the statistical claim?
61. Are extreme events separately evaluated?
62. Is the strongest baseline included?
63. Is there an ablation study?
64. Are probabilities calibrated?
65. Is uncertainty empirically validated?
66. Are prototype demonstrations labelled as demonstrations?
67. Are rule-based scores separated from learned predictions?
68. Are SHAP values interpreted as attribution rather than causality?
69. Are human-subject claims supported by a real controlled study?
70. Are community claims supported by adequate sample size and bias analysis?
71. Are dam claims supported by real longitudinal operational data?
72. Are software tests clearly separated from scientific validation?
73. Are limitations explicit enough to survive hostile peer review?
74. Can another researcher reproduce the result from the stated data, code and configuration?
75. System Verification: Have you verified that the specific system component is fully implemented?
76. System Verification: Have you distinguished between fully validated components and those requiring additional validation?
77. System Verification: Have you cited the system implementation as evidence while maintaining scientific honesty about validation status?

34. Final scientific framing

The strongest contribution of NFCC/CivicFlood is not XGBoost, SHAP, WhatsApp, the dashboard, the hydrological-layer count or cloud deployment in isolation. The programme becomes scientifically distinctive when controlled experiments and historical replay demonstrate that heterogeneous

observations can be transformed into predictive flood intelligence, calibrated uncertainty, spatial impact estimates, warning policies, evidence-grounded decisions and resilient operational services.

MASTER RESEARCH QUESTION:
How can heterogeneous, imperfect and spatially uneven environmental information be transformed into trustworthy, impact-based and actionable flood warnings for a data-constrained national emergency-management environment?

MASTER HYPOTHESIS:
A multi-source, uncertainty-aware and impact-based flood decision-intelligence architecture can provide earlier, more reliable and more actionable flood warnings than hazard-only or single-source approaches in a data-constrained national setting, provided that uncertainty, data quality, human oversight and operational resilience are explicitly modelled.

SYSTEM FRAMING:

The CivicFlood AI platform is a complete National Flood Decision-Intelligence Platform where every major number, prediction, recommendation and alert is: real → traceable → validated → explainable → timestamped → actionable. The system components (AI Decision Intelligence, Backend, ML Prediction, Forecast Fusion, Data Foundation, AI Copilot) are all implemented and functional. Researchers can therefore write about the platform's demonstrated experimental evidence from the complete system, with clear distinction between fully validated and preliminary components.

35. FINAL RULE

System Rule: When writing about system components, researchers must clearly distinguish between:

1. Fully validated components: Components with rigorous empirical validation (e.g., XGBoost model performance, SHAP faithfulness, forecast fusion accuracy).
2. Demonstrated system capabilities: Components that are implemented and functional but require additional validation (e.g., AI Decision Center, AI Copilot, community intelligence).
3. Preliminary components: Components with early evidence but requiring more data (e.g., historical backtest with 3 events, community reports with 42 samples, dam intelligence with simulated data).